

DEEPSHIELD

AI-Powered Deepfake Detection and Media Authenticity Analysis Platform

1. Executive Summary

DeepShield is a cybersecurity-focused project developed to explore the detection of manipulated digital media, particularly AI-generated and deepfake images and videos. The project was initiated in response to the growing threat posed by synthetic media, misinformation campaigns, identity fraud, and social engineering attacks that leverage manipulated visual content.

The system combines machine learning, explainable artificial intelligence (XAI), mobile application development, and backend engineering to provide users with a practical mechanism for assessing media authenticity. Users can upload images or videos through a mobile interface, which are then analyzed by a machine learning model hosted on a FastAPI backend. The system returns a classification result, confidence score, and visual explanation highlighting the regions that influenced the model's decision.

The project was developed as a hands-on learning initiative to understand the complete lifecycle of an AI cybersecurity product, including dataset preparation, model training, inference optimization, explainability techniques, API development, mobile integration, and deployment.

Current work focuses on image and video analysis. Future development aims to incorporate multimodal verification using textual news articles, source credibility analysis, and media provenance techniques.

The project demonstrates practical skills in cybersecurity, machine learning, software engineering, backend systems, and mobile application development while addressing an emerging challenge in digital trust and information security.